



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI INHA UNIVERSITY IN TASHKENT

STUDENT NAME:

STUDENT ID NUMBER:

(DO NOT USE.)

MIDTERM EXAMINATION — Spring 2024

COURSE NAME:

Signals and Systems

COURSE NUMBER:

CSE2090

EXAMINATION DATE:

-04-2024

TIME:

EXAMINATION DURATION:

60 minutes

ADDITIONAL MATERIALS
ALLOWED TO USE:

Calculators are NOT allowed.

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write ID number in each page of your examination paper.

All answers should be written in the space provided for each question, unless specified the other way.

If you have a problem please raise your hand and wait quietly for a Proctor.

You are not allowed to leave the exam room until you submit the exam papers.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17

18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34

[1-3] Express each of the following complex numbers in polar form, $re^{j\theta}$. Please fill in the answer table.

[1] $-1 - j$
 (1) $r=1, \theta=\pi/4$ (2) $r=\sqrt{2}, \theta=\pi/2$ (3) $r=1, \theta=-\pi/2$
 (4) $r=\sqrt{2}, \theta=\pi/4$ (5) $r=\sqrt{2}, \theta=-3\pi/4$

[2] $(1 - j)^2$
 (1) $r=\sqrt{2}, \theta=3\pi/4$ (2) $r=\sqrt{2}, \theta=\pi/2$ (3) $r=2, \theta=-\pi/2$
 (4) $r=1, \theta=-3\pi/4$ (5) $r=1, \theta=3\pi/4$

[3] $j(1 - j)$
 (1) $r=1, \theta=-\pi$ (2) $r=2, \theta=0$ (3) $r=\sqrt{2}, \theta=-\pi$
 (4) $r=\sqrt{2}, \theta=\pi/4$ (5) $r=\sqrt{2}, \theta=\pi/2$

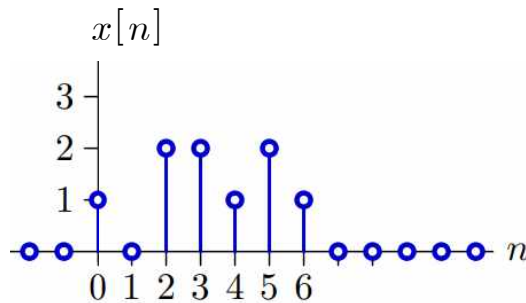
[4-6] Express each of the following complex numbers in rectangular form, $a + jb$. Please fill in the answer table.

[4] $\sqrt{2} e^{j\pi/4}$
 (1) $1+j$ (2) $-1+j$ (3) $-1-j$ (4) $1-j$ (5) $\sqrt{2}+j$

[5] $\{\sqrt{2} e^{j\pi/4}\} \cdot \{e^{-j(7\pi/2)}\}$
 (1) $1+j$ (2) $-1+j$ (3) $\sqrt{2}+j$ (4) $-1+j\sqrt{2}$ (5) $\sqrt{2}-j\sqrt{2}$

[6] $e^{j\pi/4} + e^{-j\pi/4}$
 (1) 1 (2) $\sqrt{2}$ (3) $-1+j$ (4) $-j\sqrt{2}$ (5) $j\sqrt{2}$

[7-15] A discrete-time signal $x[n]$ is shown below. When $y[n] = (x[5-n] + x[n])u[n-1]$, find the values of $y[n]$ and fill in the answer table.



- [7] $y[-2]$
- [8] $y[-1]$
- [9] $y[0]$
- [10] $y[1]$
- [11] $y[2]$
- [12] $y[3]$
- [13] $y[4]$
- [14] $y[5]$
- [15] $y[6]$

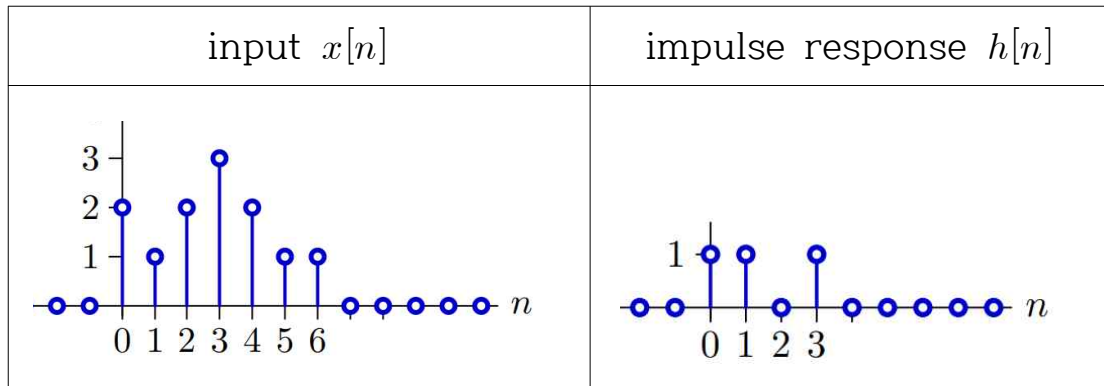
[16-20] Consider a continuous-time signal $x(t) = \cos(\omega_a t)$.

$x(t)$ is converted to the discrete-time signal $x[n] = \cos(\omega_d n)$ with the sampling frequency f_s . Find the digital frequency ω_d [rad] in each case, and fill in the answer table.

- [16] $\omega_a = 2,000 \pi$ [rad/sec] , $f_s = 8$ [kHz]
- [17] $\omega_a = 2,000 \pi$ [rad/sec] , $f_s = 4$ [kHz]
- [18] $\omega_a = 4,000 \pi$ [rad/sec] , $f_s = 8$ [kHz]
- [19] $\omega_a = 8,000 \pi$ [rad/sec] , $f_s = 8$ [kHz]
- [20] $\omega_a = 4,000 \pi$ [rad/sec] , $f_s = 1$ [kHz]

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[21-27] Consider an LTI system with impulse response $h[n]$ and input $x[n]$. Find the output $y[n]$ and fill in the answer table.



[21] $y[-2]$

[22] $y[0]$

[23] $y[2]$

[24] $y[4]$

[25] $y[6]$

[26] $y[8]$

[27] $y[10]$

[28-34] When $x[n] = \delta[n] + 2\delta[n-1] - \delta[n-3]$ is the input to an LTI system and $h[n] = 2\delta[n+1] - 2\delta[n-1]$ is the impulse response of the system. Compute output $y[n] = x[n] * h[n]$ and fill in the answer table.

[28] $y[-1]$

[29] $y[0]$

[30] $y[1]$

[31] $y[2]$

[32] $y[3]$

[33] $y[4]$

[34] This system is a causal system. True[T] or False[F]?